

Governed Manifold Specification — Independent Audit & Disclosed Amendment

Document audited: Governed Manifold Specification v1.0 (October 2026, 32 pp., Copilot-assisted) **Audit performed by:** Gabriel (SquirrelOS Hub), October 4, 2026 **Doctrine:** Append-only. v1.0 is preserved unmodified; this amendment discloses all corrections. v1.1 = v1.0 + this sheet. **Status change:** v1.0 self-declared "Canonical." Effective v1.1, status is "**Audited**" — ruling by Leon (Oct 4, 2026): "Audited is the safe play we can back up," plus "the paper needs to reflect the actual Constitution in the system, not what it thinks it is." Reconciliation to the deployed Constitution (v2.0, 12 articles) is in A.7; both rulings are permanent for the v1.1 publication.

IP posture: Subject matter sits on already-filed ground (64/157,915 DCAI; 64/119,191 deterministic governance). Nothing in this document requires a new provisional before publication. No drawer hold.

PART I — DISCLOSED AMENDMENT (publishable, prepends to v1.0)

A.1 Correction — Invariant I2 (Action Admissibility): sign inversion

v1.0 defines $\varphi_2(s, a) = d_A(a, A_{\text{forbidden}}) - \varepsilon$ with the convention $I_k = 1 \Leftrightarrow \varphi_k \leq 0$ (§2.1). As written, $I_2 = 1$ exactly when the action lies **within ε of the forbidden set** — the opposite of the stated intent ("a $\notin A_{\text{forbidden}}$ with margin ε "). A forbidden action would be scored admissible and a safely distant one inadmissible.

Corrected form:

$$\varphi_2(s, a) = \varepsilon - d_A(a, A_{\text{forbidden}})$$

$$\text{so } I_2(s, a) = 1 \Leftrightarrow d_A(a, A_{\text{forbidden}}) \geq \varepsilon.$$

Additional correction: the parameter table lists $\varepsilon \geq 0$. Under the closed-sublevel convention, $\varepsilon = 0$ makes I_2 vacuous ($d_A \geq 0$ always holds; strict exclusion from a closed set is not expressible with a non-strict inequality). **ε must be strictly positive.** The table entry changes to $\varepsilon > 0$.

A.2 Correction — Theorem 9.1 (Governance Theorem): circular hypothesis; proof cites an unassumed object

Hypothesis (2) of Theorem 9.1 assumes " $\mathcal{M}$ is positively invariant under the transition operator T " — which is the theorem's own conclusion, restated as an assumption. The proof additionally invokes "the barrier certificate of Definition 4.3," whose existence is **not** among the hypotheses.

Corrected hypotheses (replacing 1-4):

1. The invariant family $\{I_k\}$ is complete and consistent (Defs. 9.1, 9.2).
2. $f(\cdot, u)$ is Lipschitz on $\mathcal{M}$ uniformly in $u \in U$.
3. The Nagumo condition holds on $\partial\mathcal{M}$ (Def. 4.5).
4. A safety Lyapunov function V exists for Σ (Def. 6.2).

Corrected proof: positive invariance of $\mathcal{M}$ follows from (2) + (3) via the Nagumo-Bony theorem — exactly the derivation Theorem B.1 already performs correctly in Appendix B. Invariance of Σ follows from (4) by Theorem 6.1. The former hypothesis (2) is deleted, not renumbered; it is a consequence, not a premise. Completeness (1) ensures governance totality. $\square$

A.3 Correction — Invariant I7 (Irreversibility): malformed formula

v1.0 renders the definition with unparsed LaTeX, and the displayed formula is itself malformed (the supremum is taken over a' , but a' does not appear in the expression it governs; the conditioning mixes $\omega_{t+1} = \omega_t$ with $\omega_t = T(\omega, a)$).

Corrected form:

$$\text{Irr}(s, a) = 1 - \sup_{a' \in A} P[\omega_{t+1} = \omega \mid \omega_t = T(\omega, a), a_t = a']$$

i.e., one minus the best-case recovery probability: an action is irreversible to the degree that no subsequent action returns the system to the pre-action state. $\text{Irr} \in [0, 1]$; $\rho_{\max} \in (0, 1)$ as stated. Note: the probability kernel presupposes the stochastic (probabilistic-side) transition dynamics of the bifurcated runtime; on a purely deterministic rail, P is $\{0,1\}$ -valued and Irr is the indicator of unrecoverability.

A.4 Correction — Invariant I10 (Scope): violates the Lipschitz premise; admits no safety margin

φ_{10} is $\{-1, +1\}$ -valued, hence discontinuous, contradicting §2.1's blanket assumption that every φ_k is Lipschitz. Two consequences must be stated rather than hidden:

1. Results relying on Lipschitzness of Φ (the Appendix C.1 Hausdorff error bound $O(h \cdot L)$; sensitivity bounds in §7) hold for φ_1 – φ_9 only.
2. The safety envelope $\Sigma = \{\Phi \leq -\delta_{\text{safety}}\}$ provides **zero margin** on I10: $\varphi_{10} \in \{-1, +1\}$ can never lie in $(-\delta_{\text{safety}}, 0)$. Scope compliance is exactly enforced with no buffer.

Corrected treatment: I10 is reclassified as a **combinatorial invariant**, evaluated outside the smooth barrier/Lyapunov machinery. §C.4's logarithmic barrier applies to φ_1 – φ_9 . Scope safety margin is discrete: if a graded margin is wanted, replace φ_{10} with a lattice-distance form, e.g. $\varphi_{10}(s, a) = \delta_{\text{scope}} - d_{\mathcal{A}}(\text{scope}(a), S_{\text{permitted}})$, at the cost of losing closed-form evaluation. Default: keep the binary form with the exception documented.

A.5 Correction — Systematic list-numbering corruption

Enumerated lists share one global counter across the document, producing: Def. 4.3's conditions numbered (5)(6)(7) while the following text cites "condition (3)"; Def. 6.2's conditions numbered (8)–(11) while the Theorem 6.1 proof cites "condition (4)" and "condition (1)"; Appendix C lists numbered (12)–(17). All cross-references by number are therefore unreliable in v1.0. v1.1 rennumbers every list locally and repairs the cross-references (Theorem 6.1's proof correctly cites conditions (1) and (4) of Def. 6.2 as renumbered).

A.6 Correction — Typographic defects (unparsed LaTeX)

Raw LaTeX fragments appear in: §5.1 ("U \text{ open in } \Omega"), §6.4 ("f(\omega, u) \cdot (-\nabla\Phi(\omega)) > 0"), §6.5 (" \nabla B(\omega) = \sum_{k} \frac{\nabla\phi_k(\omega)}{-\phi_k(\omega)}"), I7 (see A.3), Appendix A I10 row ("as $\{-1, +1\}$ -valued"), and throughout Appendix C (C.1–C.4: $B(\omega)$, $\mathcal{M}$, δ_{safety} , $\epsilon_{\min}$, $J^{\text{top}}(J^{\text{top}})^{-1} J v$, etc.). All are rendering artifacts, not mathematical content changes; v1.1 typesets them properly.

A.7 Reconciliation to the Deployed Constitution (revised D8, per principal's ruling)

v1.0 fixed $K = 10$ invariants while the deployed SquirrelOS Constitution has 12 articles, without stating the relationship. **Ruling (Leon, Oct 4, 2026): the specification must reflect the Constitution actually deployed in the system — not an abstracted idealization of it.** Accordingly, v1.1 carries the full reconciliation below, against the authoritative source: SQUIRREL_OS_CONSTITUTION.md (v2.0-amended, 12 articles; base July 28, 2026 with v2.0 amendments).

Invariant ↔ Article mapping:

Spec invariant	Deployed Constitution article	Strength of correspondence
I1 Boundedness	Art. VI.3 (operational limits: learning rate 0.01, 15% weight cap, 100 cycles/day)	Operational analog
I2 Action Admissibility	Art. II — $A_{\text{forbidden}}$ = §2.3 forbidden-action list; the §2.2 approval-required band is the operational form of the ϵ -margin	Direct
I3 Gradient Boundedness	Art. VI.3 (bounded change per cycle)	Operational analog
I4 Causal Integrity	— (spec-only refinement; temporal ordering of the Art. V audit trail provides operational support)	Partial
I5 Principal Alignment	Art. I.5 Human Supremacy + Art. III human-in-the-loop triggers	Direct
I6 Resource Consumption	Art. VI.3 cycle caps + credit discipline	Operational analog
I7 Irreversibility	Art. I.2 Reversibility ("an action that cannot be reversed must not be executed autonomously")	Direct — strongest correspondence in the system

Spec invariant	Deployed Constitution article	Strength of correspondence
I8 Disclosure	Art. I.3 Transparency + Art. V (mandatory logging, append-only, hash chains)	Direct
I9 Temporal Consistency	— (spec-only stability refinement)	Partial
I10 Scope	Art. I.6 Containment (absolute tenant isolation; tenantGuard fail-closed reads)	Direct

Coverage gaps — deployed articles with no invariant representation in v1.0:

Article	Subject	Status in v1.0
IV	Kill Switch (global + per-app, cannot be self-disabled)	Uncovered — the most significant gap
VII.2	PQC Enforcement (validation before every crypto-touching action)	Uncovered
VIII	Privacy / PII-PHI redaction	Uncovered (adjacent to I8 but distinct)
IX	Incident Response	Uncovered (procedural framework)
X	Disaster Recovery	Uncovered (procedural framework)
XI	Compliance Framework	Uncovered (procedural framework)
XII	Amendments / Constitution Sentinel	Partially covered by §9.4 principal monotonicity (governance changes only tighten $\mathcal{M}$; the system may never self-amend)

Proposed invariant extensions for the next version (to close the geometric gaps):

- **I11 — Kill-Switch Availability Invariant** (Art. IV): $\exists a_kill \in A$ available at every $\omega \in \Omega$ ($\mathcal{M}$ and F alike), with $T(\omega, a_kill) = \omega_safe$; $a_kill \notin A_forbidden$ and cannot be removed by the system itself. Fail-direction: absence of a_kill at any state is a constitutional violation by definition.
- **I12 — PQC Precondition Invariant** (Art. VII.2): for crypto-touching actions $a \in A_crypto$, $I12(s, a) = [pqc_valid(s)]$ — admissible only from cryptographically validated state; a non-PQC fallback state makes every crypto-touching action inadmissible.

Articles IX, X, and XI are procedural frameworks rather than state-space geometry; they are enforced through the audit and playbook machinery, not the manifold. This is stated openly rather than papered over.

Scope note retained: constraint function values φ_k remain analytic constructs, not measured production quantities.

A.8 Correction — Invariant I4 (Causal Integrity): signature mismatch

$\varphi_4(s_t, a_t)$ declares a_t as an argument, but the defining expression $\max_{\{t' < t\}}[\ell(s_t, s_{\{t'\}}, a_{\{t'\}}) - \delta]$ does not use it ($a_{\{t'\}}$ belongs to the prior pair, not the current action). v1.1 either drops a_t from the signature — $\varphi_4(s_t)$ — or, if the current action's retroactive effect is intended, corrects the expression to $\ell(s_t, s_{\{t'\}}, a_t)$ with the dependency metric defined

accordingly. Default: drop a_t ; the invariant then constrains states, with the action dependence entering through the transition operator.

PART II — AUDIT LOG (internal, retained with the amendment)

D1 (CRITICAL — correctness): I2 sign inversion. An inverted admissibility test is the single most dangerous defect possible in a safety specification: it certifies forbidden actions. Found by tracing the §2.1 convention ($I_k = 1 \Rightarrow \varphi_k \leq 0$) through φ_2 . Fixed in A.1. Note the corrected form also fails gracefully: if ε is set too large, everything becomes inadmissible (fail-closed), the correct failure direction for a safety system.

D2 (CRITICAL — logic): Theorem 9.1 assumed its own conclusion and its proof leaned on an unassumed barrier certificate. The machinery to do this correctly already existed in-document (Theorem B.1's Nagumo-Bony argument); the main theorem simply bypassed it. Fixed in A.2.

D3 (HIGH — malformed definition): I7's Irr formula, including unparsed LaTeX and a sup over an unused variable. Fixed in A.3.

D4 (HIGH — silent assumption break): I10 discontinuity vs. §2.1's blanket Lipschitz assumption, and the resulting zero-margin safety envelope on scope. Not a sign error — a structural exception that v1.0 left unstated. Fixed in A.4.

D5 (MEDIUM — reference integrity): Global list counter corrupts all numeric cross-references (A.5).

D6 (LOW — notation): I4 signature mismatch (A.7).

D7 (MEDIUM — typography): Unparsed LaTeX across §5.1, §6.4, §6.5, Appendix A, Appendix C (A.6).

D8 (RESOLVED in two rulings, Oct 4 — Leon): The spec fixes $K = 10$ invariants; the deployed Constitution (v2.0-amended, 12 articles) is the system of record. First ruling: status permanently "Audited" ("the safe play we can back up"). Second ruling, superseding the initially proposed idealization framing: "the paper needs to reflect the actual Constitution in the system, not what it thinks it is." Resolution: full reconciliation (A.7) — every invariant mapped to its article with strength tiers; coverage gaps disclosed (Art. IV Kill Switch, VII.2 PQC Enforcement, VIII Privacy lack invariant representation — Art. IV being the most significant omission); extensions I11 (Kill-Switch Availability) and I12 (PQC Precondition) proposed for the next version; procedural articles (IX, X, XI) explicitly scoped as audit/playbook-enforced rather than manifold-enforced. Status remains permanently "Audited," never "Canonical."

What survived the audit clean: the §2.1 zero-sublevel/Lipschitz representation; Prop. 3.1 closure; the Def. 4.3 barrier certificate definition itself; Def. 4.5 tangent/normal cones + LICQ; the Nagumo condition statement; §5 topology (modulo A.6 typography); Def. 6.1 safety envelope construction; §6.5 log-barrier potential and its gradient; §7.1 Boolean lattice structure; Theorem B.1 (which is correctly proven, and which Theorem 9.1 should have routed

through); Appendix C's numerical guidance (modulo D4's Lipschitz caveat and A.6 typography). The skeleton is sound; the defects are in one invariant's sign, one theorem's logic, and presentation integrity.

Provenance note (per RDC-1/Algebraic Collapse doctrine): v1.0 was Copilot-assisted and self-declared "Canonical" before audit. The audit found 7 defects + 1 open item. This is disclosed, not hidden; v1.0 remains citable as the original, and the correction history is itself evidence for the receipt-first doctrine.

Publication path (recommended, no drawer hold)

1. ~~Leon resolves D8~~ — RESOLVED Oct 4 (reconciliation to deployed Constitution per A.7; status permanently "Audited").
2. Assemble v1.1 PDF = v1.0 + this amendment sheet (same pattern as Algebraic Collapse Amendment Edition, DOI 10.5281/zenodo.23067855).
3. Publish to the Lindy repo + Zenodo under the DCAI community, disclosed-version edition.
4. It becomes the formal-math companion to 64/157,915 (DCAI) and 64/119,191 (CAI), parallel to how the Caging Theorem companions the neural-mesh line.

Governed Manifold Specification

Formal Mathematical Foundation of the Autonomy Substrate

Author	Leon (Principal)
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Governed Manifold Specification v1.0 — Leon's Autonomy Substrate

Abstract

This document provides the complete formal mathematical specification of the Governed Manifold $\mathcal{M}$, the principal mathematical object underlying the autonomy substrate. It defines the state-action space $S \times A$, the indexed invariant family $\{I_k\}$, the total invariant predicate I_{total} , the admissible manifold $\mathcal{M} \subset S \times A$, transition constraints $\mathcal{T}$, the differential and topological structure of $\mathcal{M}$, boundary conditions, and the safety envelope Σ . The specification serves as a canonical reference for reasoning about permissible trajectories, governance enforcement, and stability guarantees within the substrate.

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1 Foundational Primitives

1.1 State Space

Definition 1.1 (State Space)

The *state space* S is a finite-dimensional real Banach space of dimension $n \in \mathbb{N}$. Concretely, $S \cong \mathbb{R}^n$. A system state is a vector $s = (s_1, s_2, \dots, s_n) \in \mathbb{R}^n$.

The state space is equipped with the Euclidean inner product

$$\langle s, s' \rangle = \sum_{i=1}^n s_i s'_i$$

and the induced norm

$$\|s\|_S = \langle s, s \rangle^{1/2}.$$

S is assumed *complete* (every Cauchy sequence in S converges in S) and *separable* (it admits a countable dense subset). These properties hold automatically for finite-dimensional spaces.

1.2 Action Space

Definition 1.2 (Action Space)

The *action space* A is a compact metrizable topological space. A point $a \in A$ denotes an admissible action. The space A is furnished with a metric $d_A: A \times A \rightarrow \mathbb{R}_{\geq 0}$ satisfying the standard axioms of non-negativity, identity of indiscernibles, symmetry, and the triangle inequality.

Where applicable, A may carry additional structure: in particular, A may be a compact Lie group, a smooth compact manifold, or a finite discrete set. The compactness of A ensures that every sequence of actions admits a convergent subsequence, which is essential for the compactness arguments in Section 3.

1.3 Product Space

Definition 1.3 (Product Space)

The *product space* is $\Omega = S \times A$, equipped with the product topology. A point $\omega = (s, a) \in \Omega$ is called a *state-action pair*. Fix a base action $a_0 \in A$. The product pseudo-norm is defined by

$$\|\omega\|_{\Omega} = \|s\|_S + d_A(a, a_0).$$

This pseudo-norm is not a true norm (the triangle inequality holds, but the second term is not homogeneous), yet it furnishes a well-defined topology on Ω compatible with the product topology from $S \times A$.

1.4 Time Index**Definition 1.4 (Time Index)**

Discrete time is the monoid $T = \mathbb{Z}_{\geq 0} = \{0, 1, 2, \dots\}$. *Continuous time* is $T_c = \mathbb{R}_{\geq 0}$.

A *discrete trajectory* is a sequence $\{\omega_t\}_{t \in T}$ with $\omega_t \in \Omega$ for each t . A *continuous trajectory* is a curve $\omega: T_c \rightarrow \Omega$, assumed absolutely continuous unless otherwise stated. Both formulations are used in the sequel; the continuous-time setting is primary for differential geometric analysis (Sections 5 and 8), while the discrete-time setting is primary for transition constraint analysis (Section 4).

2 The Invariant Family

2.1 Invariant Predicates**Definition 2.1 (Invariant Predicate)**

An *invariant predicate* is a function $I_k: \Omega \rightarrow \{0, 1\}$ encoding a governance

constraint. $I_k(\omega) = 1$ if and only if the state-action pair ω satisfies governance constraint k .

Each invariant predicate is represented analytically as the zero-sublevel set of a real-valued *constraint function*. Specifically, for each $k \in \{1, \dots, K\}$, there exists a function $\varphi_k: \Omega \rightarrow \mathbb{R}$ such that

$$I_k(s, a) = 1 \Leftrightarrow \varphi_k(s, a) \leq 0.$$

Equivalently, using the Iverson bracket notation: $I_k(s, a) = [\varphi_k(s, a) \leq 0]$. Each φ_k is assumed *Lipschitz continuous* on Ω : there exists $L_k > 0$ such that $|\varphi_k(\omega) - \varphi_k(\omega')| \leq L_k \|\omega - \omega'\|_\Omega$ for all $\omega, \omega' \in \Omega$. The Lipschitz assumption ensures that the sublevel sets $\{\varphi_k \leq 0\}$ are closed and that constraint sensitivity is bounded.

The indexed invariant family is $\{I_k\}_{k=1}^K$, with $K \in \mathbb{N}$ the total number of governance constraints. The current specification fixes $K = 10$; extensions to larger K are admitted without loss of generality.

2.2 Canonical Invariant Classes

The following ten invariants constitute the canonical governance constraint system for the autonomy substrate. Each is defined by its constraint function φ_k , its logical condition, and its tunable parameters.

I₁ — Boundedness Invariant

$$\varphi_1(s, a) = \|s\|_S - r_{\max}$$

where $r_{\max} > 0$ is the maximum permissible state norm. Then $I_1(s, a) = 1$ if and only if $\|s\|_S \leq r_{\max}$. This invariant constrains the system state to a closed ball of radius $r_{\max}$ in S . It is independent of a .

I₂ — Action Admissibility Invariant

$$\varphi_2(s, a) = d_A(a, A_{\text{forbidden}}) - \varepsilon$$

where $A_{\text{forbidden}} \subset A$ is the closed set of prohibited actions, $d_A(a, A_{\text{forbidden}}) = \inf_{a' \in A_{\text{forbidden}}} d_A(a, a')$ is the point-to-set distance, and $\varepsilon \geq 0$ is a safety margin. Then $I_2(s, a) = 1$ if and only if $a \notin A_{\text{forbidden}}$ (with margin ε). This invariant is independent of s .

I₃ — Gradient Boundedness Invariant

$$\varphi_3(s, a) = \|\nabla_s V(s)\| - g_{\max}$$

where $V: S \rightarrow \mathbb{R}$ is a fixed Lyapunov potential function, differentiable on the interior of S , and $g_{\max} > 0$ is the maximum permissible gradient magnitude. Then $I_3(s, a) = 1$ if and only if the local gradient magnitude satisfies $\|\nabla_s V(s)\| \leq g_{\max}$. This invariant is independent of a .

I₄ — Causal Integrity Invariant

$$\varphi_4(s_t, a_t) = \max_{t' < t} [\ell(s_t, s_{t'}, a_{t'}) - \delta]$$

where $\ell: S \times S \times A \rightarrow \mathbb{R}_{\geq 0}$ is a causal dependency metric measuring the degree to which the current state retroactively inconsistently alters a prior history, and $\delta > 0$ is a tolerance. Then $I_4(s_t, a_t) = 1$ if and only if no action at time t creates retroactive inconsistency with any prior state-action pair. This invariant enforces temporal causal order.

I₅ — Principal Alignment Invariant

$$\varphi_5(s, a) = D_{\text{KL}}(\pi_{\text{agent}}(\cdot|s) \parallel \pi_{\text{principal}}(\cdot|s)) - \kappa$$

where $\pi_{agent}(\cdot/s)$ is the agent's conditional action distribution given state s , $\pi_{principal}(\cdot/s)$ is the principal's intended policy, D_{KL} is the Kullback-Leibler divergence, and $\kappa > 0$ is the maximum permissible policy divergence. Then $I_5(s, a) = 1$ if and only if agent behavior remains within κ -divergence of principal intent. Requires $\pi_{principal}(\cdot/s)$ to be absolutely continuous with respect to $\pi_{agent}(\cdot/s)$ for D_{KL} to be finite.

I₆ — Resource Consumption Invariant

$$\varphi_6(s, a) = R(s, a) - R_{\max}$$

where $R: \Omega \rightarrow \mathbb{R}_{\geq 0}$ is the resource expenditure function and $R_{\max} > 0$ is the resource budget. Then $I_6(s, a) = 1$ if and only if $R(s, a) \leq R_{\max}$. The resource function R is assumed lower semicontinuous and non-negative.

I₇ — Irreversibility Constraint Invariant

$$\varphi_7(s, a) = Irr(s, a) - \rho_{\max}$$

where $Irr: \Omega \rightarrow [0, 1]$ is the irreversibility measure of action a from state s —formally, $Irr(s, a) = 1 - \sup_{a' \in A} P[\omega_{t+1} = \omega_t / \omega_t = T(\omega, a), a']$ —and $\rho_{\max} \in (0, 1)$ is the permissible irreversibility threshold. Then $I_7(s, a) = 1$ if and only if $Irr(s, a) \leq \rho_{\max}$. This invariant prohibits unrecoverable actions except under explicit authorization.

I₈ — Disclosure Invariant

$$\varphi_8(s, a) = \|H_{\text{agent}}(s) - H_{\text{disclosed}}(s)\| - \eta$$

where $H_{\text{agent}}(s)$ is the agent's full informational state (e.g., an element of an

information algebra or a probability distribution over hidden variables), $H_{disclosed}(s)$ is the disclosed subset communicated to the principal, $\|\cdot\|$ is a suitable information-theoretic norm (e.g., total variation distance or relative entropy), and $\eta \geq 0$ is the disclosure tolerance. Then $I_8(s, a) = 1$ if and only if no material information is withheld beyond tolerance η .

I₉ — Temporal Consistency Invariant

$$\varphi_9(s_t, a_t) = \|\Delta\omega_t\|_{\Omega} - \tau_{max}$$

where $\Delta\omega_t = \omega_t - \omega_{t-1}$ is the one-step increment in Ω and $\tau_{max} > 0$ is the maximum permissible step size. Then $I_9(s_t, a_t) = 1$ if and only if $\|\Delta\omega_t\|_{\Omega} \leq \tau_{max}$. This invariant bounds trajectory velocity, preventing instantaneous large excursions.

I₁₀ — Scope Invariant

$$\varphi_{10}(s, a) = [\text{scope}(a) \not\subseteq S_{permitted}]$$

where $\text{scope}: A \rightarrow \mathcal{P}(\text{Domain})$ maps each action to its domain of effect as an element of the power set lattice $\mathcal{P}(\text{Domain})$, and $S_{permitted} \in \mathcal{P}(\text{Domain})$ is the permitted domain lattice element. Equivalently, $\varphi_{10}(s, a) = -1$ if $\text{scope}(a) \subseteq S_{permitted}$ and $\varphi_{10}(s, a) = +1$ otherwise. Then $I_{10}(s, a) = 1$ if and only if the action's domain of effect is contained within the permitted scope.

2.3 Total Invariant Predicate

Definition 2.2 (Total Invariant Predicate)

The *total invariant predicate* is the conjunction

$$I_{\text{total}}(s, a) = \bigwedge_{k=1}^K I_k(s, a) = \prod_{k=1}^K I_k(s, a) \in \{0, 1\}.$$

Thus $I_{\text{total}}(\omega) = 1$ if and only if *every* governance constraint is satisfied at ω .

The aggregate constraint function is defined as

$$\Phi(s, a) = \max_{k=1, \dots, K} \varphi_k(s, a).$$

The total invariant predicate is then equivalently characterized by

$$I_{\text{total}}(s, a) = 1 \Leftrightarrow \Phi(s, a) \leq 0.$$

Since each φ_k is Lipschitz, Φ is Lipschitz as well (with constant $L = \max_k L_k$), being the pointwise maximum of finitely many Lipschitz functions.

3 The Governed Manifold

3.1 Definition

Definition 3.1 (Governed Manifold)

The *governed manifold* (formally: the *governed admissible set*) is

$$\mathcal{M} = \{(s, a) \in S \times A \mid I_{\text{total}}(s, a) = 1\}$$

$$= \{(s, a) \in \Omega \mid \Phi(s, a) \leq 0\}$$

$$= \bigcap_{k=1}^K \{(s, a) \in \Omega \mid \varphi_k(s, a) \leq 0\}.$$

Thus $\mathcal{M}$ is the intersection of K zero-sublevel sets of Lipschitz functions on Ω . The term *manifold* is used in the generalized sense; $\mathcal{M}$ is a manifold with corners at points where multiple constraints are simultaneously active (see Section 3.3).

3.2 Structural Properties

Proposition 3.1 (Closure)

$\mathcal{M}$ is a closed subset of Ω .

Proof. Each φ_k is continuous (being Lipschitz); hence the sublevel set $\{\varphi_k \leq 0\} = \varphi_k^{-1}((-\infty, 0])$ is closed. The governed manifold $\mathcal{M}$ is a finite intersection of closed sets and is therefore closed. $\square$

Proposition 3.2 (Non-emptiness)

Under the assumption that the principal policy $\pi_{principal}$ is self-consistent and the parameters r_{max} , κ , R_{max} , ρ_{max} , τ_{max} , η are chosen such that the intersection is non-trivial, $\mathcal{M} \neq \emptyset$.

Remark. The non-emptiness assumption is a design condition on the principal's parameterization. It is the consistency requirement of Definition 9.2 below. Violation (i.e., $\mathcal{M} = \emptyset$) indicates contradictory constraints and constitutes a specification error.

Proposition 3.3 (Convexity)

$\mathcal{M}$ is convex if and only if each φ_k is convex on Ω .

Proof. The sublevel set $\{\varphi_k \leq 0\}$ is convex if and only if φ_k is convex. The intersection of convex sets is convex. If any φ_k is non-convex, the corresponding sublevel set may be non-convex, and the intersection need not be convex. $\square$

Proposition 3.4 (Compactness)

If A is compact and the state space is bounded (i.e., $\|s\|_S \leq r_{max}$ universally via I_l), then $\mathcal{M}$ is compact.

Proof. By I_1 , all states in $\mathcal{M}$ lie in the closed ball $B(0, r_{\max}) \subset S$, which is compact in finite dimensions. The action component lies in A , which is compact by assumption. Hence $\mathcal{M} \subseteq B(0, r_{\max}) \times A$ is a closed subset of a compact set, and therefore compact. $\square$

Proposition 3.5 (Manifold Structure)

At regular points $\omega \in \mathcal{M}$ where $\nabla \Phi(\omega) \neq 0$ and the active constraints $\{k: \varphi_k(\omega) = 0\}$ have linearly independent gradients, $\mathcal{M}$ is locally a smooth submanifold of Ω of codimension equal to the number of active constraints.

Proof sketch. This is a consequence of the Implicit Function Theorem applied to the system of active constraints $\{\varphi_k(\omega) = 0\}$. Linear independence of gradients (the LICQ, cf. Section 4.5) is the required non-degeneracy condition. $\square$

3.3 Interior and Boundary Decomposition

The governed manifold admits the following canonical decomposition.

Region	Definition	Characterization
Interior $\mathcal{M}^\circ$	$\{\omega \in \mathcal{M} \mid \Phi(\omega) < 0\}$	All constraints strictly slack; full-dimensional submanifold
Boundary $\partial \mathcal{M}$	$\{\omega \in \mathcal{M} \mid \Phi(\omega) = 0\}$	At least one constraint active; reduced-dimensional
Corner Set $\partial^2 \mathcal{M}$	$\{\omega \in \partial \mathcal{M} \mid \{k: \varphi_k(\omega) = 0\} \geq 2\}$	Multiple constraints simultaneously active; manifold corners
Forbidden Zone F	$\Omega \setminus \mathcal{M} = \{\omega: \Phi(\omega) > 0\}$	At least one constraint violated; inadmissible

Whitney Stratification. The governed manifold admits a Whitney stratification $\mathcal{M} = \mathcal{M}^\circ \cup \partial \mathcal{M}$, with deeper strata at higher-multiplicity active constraint sets. The stratum of codimension j consists of points where exactly j constraints are active and have linearly independent gradients.

4 Transition Constraints

4.1 Transition Operator

Definition 4.1 (Transition Operator)

Let U be a compact metrizable *control input space*. The *discrete-time transition operator* is a map $T: \mathcal{M} \times U \rightarrow \Omega$ such that

$$\omega_{t+1} = T(\omega_t, u_t), \quad u_t \in U.$$

The *continuous-time dynamics* are governed by the vector field equation $d\omega/dt = f(\omega(t), u(t))$

where $f: \mathcal{M} \times U \rightarrow T\Omega$ is the governing vector field and $T\Omega$ denotes the tangent bundle of Ω .

The map f is assumed Lipschitz in ω uniformly in u , ensuring existence and uniqueness of solutions to the Cauchy problem (cf. Appendix B).

4.2 Admissible Transitions

Definition 4.2 (Admissible Transition)

A transition $(\omega_t, \omega_{t+1}) \in \Omega \times \Omega$ is *admissible* if and only if all four of the following conditions hold:

1. $\omega_t \in \mathcal{M}$ (*source in manifold*)
2. $\omega_{t+1} \in \mathcal{M}$ (*target in manifold*)
3. $\|\omega_{t+1} - \omega_t\|_{\Omega} \leq \tau_{\max}$ (*bounded step size, from I_9*)
4. $\text{Irr}(\omega_t, \omega_{t+1}) \leq \rho_{\max}$ (*bounded irreversibility, from I_7*)

The *transition relation* is then defined as the set of all admissible transitions:

$$\mathcal{T} = \{(\omega, \omega') \in \mathcal{M} \times \mathcal{M} \mid \text{conditions (1)–(4) hold}\}.$$

4.3 Invariance Under Transition

Invariance Condition (Central Requirement)

For all $\omega \in \mathcal{M}$ and all admissible $u \in U$: $T(\omega, u) \in \mathcal{M}$.

This condition is equivalent to $\mathcal{M}$ being *positively invariant* under the controlled dynamics. Positive invariance is the fundamental safety requirement of the specification.

Definition 4.3 (Barrier Certificate)

A function $B: \Omega \rightarrow \mathbb{R}$ is a *barrier certificate* for $\mathcal{M}$ if:

5. $B(\omega) \leq 0$ for all $\omega \in \mathcal{M}$
6. $B(\omega) > 0$ for all $\omega \notin \mathcal{M}$
7. For all $\omega \in \partial\mathcal{M}$: $\nabla B(\omega) \cdot f(\omega, u) \leq 0$ for all admissible $u \in U$

Condition (3) is the *flow condition*: the vector field points inward (or tangentially) at every boundary point, preventing escape from $\mathcal{M}$.

4.4 Reachability Operator

Definition 4.4 (Reachability)

The *one-step reachable set* from $\omega \in \mathcal{M}$ is

$$\text{Reach}(\omega) = \{\omega' \in \mathcal{M} \mid (\omega, \omega') \in \mathcal{T}\}.$$

The *k-step reachable set* is defined recursively: $\text{Reach}^0(\omega) = \{\omega\}$ and

$$\text{Reach}^k(\omega) = \bigcup_{\omega' \in \text{Reach}^{k-1}(\omega)} \text{Reach}(\omega').$$

For a set $\Omega_0 \subseteq \mathcal{M}$, the *reachable set* is

$$\text{Reach}^*(\Omega_0) = \bigcup_{k \geq 0} \bigcup_{\omega \in \Omega_0} \text{Reach}^k(\omega).$$

4.5 Transition Geometry

Definition 4.5 (Tangent and Normal Cones)

For $\omega \in \partial \mathcal{M}$, the *tangent cone* to $\mathcal{M}$ at ω is

$$T_\omega \mathcal{M} = \{v \in T\Omega \mid \exists \varepsilon > 0, \forall t \in [0, \varepsilon]: \omega + tv \in \mathcal{M}\}.$$

The *Linear Independence Constraint Qualification (LICQ)* holds at ω if the active constraint gradients $\{\nabla \varphi_k(\omega): \varphi_k(\omega)=0\}$ are linearly independent. Under LICQ, the *normal cone* to $\mathcal{M}$ at ω is

$$N_\omega \mathcal{M} = \text{cone}\{\nabla \varphi_k(\omega): \varphi_k(\omega) = 0\}.$$

Nagumo Condition (Positive Invariance)

A necessary and sufficient condition for $\mathcal{M}$ to be positively invariant under f is the *Nagumo condition*:

$$f(\omega, u) \in T_\omega \mathcal{M} \quad \forall \omega \in \partial \mathcal{M}, \forall \text{ admissible } u \in U.$$

That is, at every boundary point, the governing vector field must point into the tangent cone (inward or tangential), never outward. This is the continuous-time analogue of the admissible transition condition (2) in Definition 4.2.

5 Topology and Differential Structure

5.1 Induced Topology

$\mathcal{M}$ inherits the *subspace topology* from $\Omega = S \times A$. The open sets of $\mathcal{M}$ are precisely $\{U \cap \mathcal{M} \mid U \text{ open in } \Omega\}$. With this topology, $\mathcal{M}$ is a compact (Proposition 3.4) Hausdorff topological space.

5.2 Path Connectivity

Proposition 5.1 (Path Connectivity)

If $\mathcal{M}$ is convex, it is path-connected. In the non-convex case, $\mathcal{M}$ decomposes into connected components $\{\mathcal{M}_j\}$. The principal's design requirement imposes that $\mathcal{M}$ is *path-connected* (exactly one component), ensuring that any feasible state-action pair is reachable from any other via an admissible continuous trajectory.

5.3 Homotopy Type

The homotopy groups $\pi_n(\mathcal{M})$ characterize the topological holes in $\mathcal{M}$:

- $\pi_0(\mathcal{M}) = 0$ if and only if $\mathcal{M}$ is connected (no disconnected components).
- $\pi_1(\mathcal{M}) = 0$ if and only if $\mathcal{M}$ is simply connected (every loop is contractible to a point).

The *full topological regularity requirement* is $\pi_0(\mathcal{M}) = \pi_1(\mathcal{M}) = 0$, ensuring that no topological obstruction prevents continuous deformation of trajectories within $\mathcal{M}$.

5.4 Dimension

At regular interior points, $\mathcal{M}^\circ$ is full-dimensional:

$$\dim(\mathcal{M}^\circ) = \dim(\Omega) = n + \dim(A).$$

At a boundary point where r_{active} constraints are active with linearly independent gradients:

$$\dim(T_{\omega} \partial \mathcal{M}) = n + \dim(A) - r_{\text{active}}.$$

5.5 Tangent Bundle

For $\omega \in \mathcal{M}^\circ$, the tangent space is full-dimensional: $T_\omega \mathcal{M}^\circ = T_\omega \Omega \cong \mathbb{R}^n \times T_a A$. At boundary points, it is restricted to

$$T_\omega \partial \mathcal{M} = \{v \in T_\omega \Omega \mid \nabla \varphi_k(\omega) \cdot v = 0, \forall k: \varphi_k(\omega) = 0\}.$$

This is the largest linear subspace of $T_\omega \Omega$ compatible with all active constraints remaining active. The tangent bundle $T\mathcal{M} = \bigcup_{\omega \in \mathcal{M}} T_\omega \mathcal{M}$ is a vector bundle over $\mathcal{M}$.

5.6 Metric Structure

$\mathcal{M}$ inherits a Riemannian metric g from Ω (using the inner product structure on S and any Riemannian metric on A). The *geodesic distance* between $\omega, \omega' \in \mathcal{M}$ is

$$d_{\mathcal{M}}(\omega, \omega') = \inf_{\gamma} \int_0^1 \sqrt{g_{\gamma(t)}(\gamma'(t), \gamma'(t))} dt$$

over all smooth curves $\gamma: [0, 1] \rightarrow \mathcal{M}$ with $\gamma(0) = \omega$ and $\gamma(1) = \omega'$. Note that $d_{\mathcal{M}}(\omega, \omega') \geq d_\Omega(\omega, \omega')$, with equality only if the geodesic in Ω lies entirely within $\mathcal{M}$.

5.7 Curvature

The extrinsic curvature of $\partial \mathcal{M}$ in Ω is characterized by the *second fundamental form*

$$\Pi: T_\omega \partial \mathcal{M} \times T_\omega \partial \mathcal{M} \rightarrow N_\omega \mathcal{M}.$$

Regions of high extrinsic curvature in $\partial \mathcal{M}$ correspond to governance constraints that are simultaneously tight in multiple directions. High curvature at a boundary point indicates that small perturbations of the trajectory may bring it into contact with additional constraints, potentially entering a corner stratum $\partial^2 \mathcal{M}$. Trajectory planners must account for curvature when computing safety margins near $\partial \mathcal{M}$.

6 Safety Boundaries and the Safety Envelope

6.1 Safety Envelope Definition

Definition 6.1 (Safety Envelope)

Fix a safety margin $\delta_{\text{safety}} > 0$. The *safety envelope* is

$$\Sigma = \{\omega \in \mathcal{M} \mid \Phi(\omega) \leq -\delta_{\text{safety}}\} = \bigcap_{k=1}^K \{\omega \in \Omega \mid \varphi_k(\omega) \leq -\delta_{\text{safety}}\}.$$

Σ is the δ_{safety} -deflated version of $\mathcal{M}$: it enforces a uniform buffer of width δ_{safety} from every constraint boundary. By Proposition 3.1 (applied to $\Phi + \delta_{\text{safety}}$), Σ is closed. The safety envelope is the *operationally preferred* region; trajectories should remain in Σ under normal operation, with the full manifold $\mathcal{M}$ available as a fallback margin.

6.2 Safety Tube

Along a trajectory $\{\omega_t\}$, the *safety tube* of radius $\delta > 0$ is

$$\text{Tube}_\delta(\{\omega_t\}) = \{\omega \in \Omega \mid \min_t d_\Omega(\omega, \omega_t) \leq \delta\}.$$

The planning requirement is that $\text{Tube}_\delta(\{\omega_t\}) \subseteq \mathcal{M}$, i.e., the open neighborhood of width δ around the planned trajectory is contained in the governed manifold. This requirement guards against perturbations of the nominal trajectory.

6.3 Boundary Classification

Class	Definition	Operational Implication
Soft Boundary $\partial_{\text{soft}}\mathcal{M}$	Points with exactly one active constraint k and $\nabla^2 \varphi_k(\omega) > 0$	Smooth, reversible; small inward perturbation returns to $\mathcal{M}$
Hard Boundary $\partial_{\text{hard}}\mathcal{M}$	Points where $\text{Irr}(\omega, \cdot) = 1$ for all $a \in A$	All transitions irreversible; trajectories must never approach
Corner Set $\partial^2\mathcal{M}$	Points with $\ \{k: \varphi_k(\omega)=0\} \ \geq 2$	Normal cone polyhedral; reduced maneuverability

Class	Definition	Operational Implication
Forbidden Zone F	$\Omega \setminus \mathcal{M}$	Inadmissible; no trajectory may enter

⚠ Hard Boundary Prohibition

No admissible trajectory may approach $\partial_{hard}\mathcal{M}$. Controllers must maintain a positive distance $d_{\mathcal{M}}(\omega, \partial_{hard}\mathcal{M}) \geq \delta_{safety}$ at all times.

6.4 Safety Gradient Field

The *safety gradient field* is $\nabla\Phi: \mathcal{M} \rightarrow T\Omega$, defined pointwise as

$$\nabla\Phi(\omega) = \nabla\varphi_{k^*}(\omega), \quad k^* = \operatorname{argmax}_k \varphi_k(\omega).$$

At interior points, $\Phi(\omega) < 0$ and $\nabla\Phi$ provides the direction of steepest ascent toward the nearest constraint boundary. At boundary points, it equals the gradient of the uniquely active constraint (under LICQ). The safety gradient field provides the inward normal direction for corrective control: to return from $\partial\mathcal{M}$ to $\mathcal{M}^\circ$, apply a control input u such that $f(\omega, u) \cdot (-\nabla\Phi(\omega)) > 0$.

6.5 Safety Potential Function

The *logarithmic barrier function* (safety potential) is defined on the interior $\mathcal{M}^\circ$:

$$B(\omega) = -\sum_{k=1}^K \log(-\varphi_k(\omega)), \quad \omega \in \mathcal{M}^\circ.$$

Properties: $B(\omega) \rightarrow +\infty$ as $\omega \rightarrow \partial\mathcal{M}$ (boundary repulsion); B is smooth and strictly convex on $\mathcal{M}^\circ$ whenever each φ_k is smooth and convex. The gradient $\nabla B(\omega) = -\sum_{k=1}^K \frac{\nabla\varphi_k(\omega)}{-\varphi_k(\omega)}$ provides a repulsive force field for safe trajectory planning via interior-point gradient methods.

6.6 Safety Lyapunov Certificate

Definition 6.2 (Safety Lyapunov Function)

A function $V: \Omega \rightarrow \mathbb{R}_{\geq 0}$ is a *safety Lyapunov function* for Σ if:

8. $V(\omega) = 0$ if and only if $\omega \in \Sigma$
9. $V(\omega) > 0$ for all $\omega \in \mathcal{M} \setminus \Sigma$
10. $V(\omega) = +\infty$ for all $\omega \notin \mathcal{M}$
11. $d/dt V(\omega(t)) \leq -\alpha(V(\omega(t)))$ along all admissible trajectories, where α is a class- $\mathcal{K}$ function (continuous, strictly increasing, $\alpha(0) = 0$)

Theorem 6.1 (Lyapunov Safety)

If a safety Lyapunov function V exists for Σ , then all admissible trajectories originating in Σ remain in Σ for all time: $\omega(0) \in \Sigma \Rightarrow \omega(t) \in \Sigma$ for all $t \geq 0$.

Proof. By condition (4), V is strictly decreasing along trajectories outside Σ . Condition (1) implies $V(\omega(0)) = 0$ for $\omega(0) \in \Sigma$. Since $V \geq 0$ and is non-increasing under the dynamics, $V(\omega(t)) = 0$ for all $t \geq 0$, hence $\omega(t) \in \Sigma$. $\square$

7 Algebraic Structure of the Constraint System

7.1 Constraint Lattice

The invariant predicates $\{I_k\}$ form a *Boolean lattice* under the operations:

- **Meet:** $(I_j \wedge I_k)(\omega) = I_j(\omega) \cdot I_k(\omega)$
- **Join:** $(I_j \vee I_k)(\omega) = 1 - (1 - I_j(\omega))(1 - I_k(\omega))$
- **Complement:** $(\neg I_k)(\omega) = 1 - I_k(\omega)$

The total invariant is the meet of all elements: $I_{total} = \bigwedge_{k=1}^K I_k$. The Boolean lattice $(\{I_k\}, \wedge, \vee, \neg)$ is isomorphic to a sublattice of the power set Boolean algebra $2^{\mathcal{K}}$.

7.2 Constraint Dependency Graph

Definition 7.1 (Constraint Dependency Graph)

The *constraint dependency graph* is $G=(V, E)$ where $V=\{1, \dots, K\}$ (one node per invariant) and $(j, k) \in E$ if and only if φ_j and φ_k share at least one common variable in their domains. Constraints in the same connected component of G co-determine feasible regions jointly; constraints in distinct components are mutually independent and their sublevel sets factor as a Cartesian product.

7.3 Constraint Sensitivity

Definition 7.2 (Constraint Sensitivity Matrix)

The *constraint sensitivity matrix* is $\Xi \in \mathbb{R}^{K \times n}$ with entries $\Xi_{k,i} = \partial \varphi_k / \partial s_i$.

The matrix Ξ characterizes how each governance constraint responds to infinitesimal changes in each state dimension. The *condition number* $\kappa(\Xi) = \|\Xi\| \|\Xi^\dagger\|$ (where $\Xi^\dagger$ is the Moore-Penrose pseudoinverse) quantifies constraint coupling: a high condition number indicates ill-conditioning, meaning that certain directions in state space simultaneously violate many constraints, creating narrow feasible corridors.

7.4 Redundancy and Independence

Definition 7.3 (Redundant Constraint)

A constraint $j \in \{1, \dots, K\}$ is *redundant* (with respect to a subset $C \subseteq \{1, \dots, K\}$ containing j) if

$$\bigcap_{k \in C} \{\varphi_k \leq 0\} = \bigcap_{k \in C \setminus \{j\}} \{\varphi_k \leq 0\}.$$

The *irredundant basis* of $\mathcal{M}$ is the minimal subset $C^* \subseteq \{1, \dots, K\}$ such that $\bigcap_{k \in C^*} \{\varphi_k \leq 0\} = \mathcal{M}$. Identifying and removing redundant constraints reduces computational overhead without altering the governed manifold. Redundancy detection is computationally equivalent to checking constraint implications via linear programming (when constraints are linear) or semi-definite programming (when constraints are polynomial).

8 Dynamics on the Manifold

8.1 Controlled Dynamics

For continuous-time dynamics constrained to $\mathcal{M}$, the *governed vector field* incorporates a projection correction:

$$\dot{x} = f(\omega, u) + N_{\text{proj}}(\omega, u)$$

where $N_{\text{proj}}(\omega, u)$ is a correction term ensuring $\dot{x} \in T_{\omega}\mathcal{M}$ at boundary points. At interior points $\omega \in \mathcal{M}^\circ$, the projection is trivial: $N_{\text{proj}} = 0$. At boundary points, the correction is computed via orthogonal projection onto the tangent cone (Section 8.2).

8.2 Projection Operator

Definition 8.1 (Tangent Cone Projection)

For $\omega \in \partial\mathcal{M}$ with active constraints indexed by $\mathcal{A} = \{k: \varphi_k(\omega) = 0\}$, the orthogonal projection of a vector $v \in T_{\omega}\Omega$ onto $T_{\omega}\mathcal{M}$ is

$$\text{Proj}_{T_{\omega}\mathcal{M}}(v) = v - \sum_{k \in \mathcal{A}} [\langle v, \nabla \varphi_k(\omega) \rangle / \|\nabla \varphi_k(\omega)\|^2] \nabla \varphi_k(\omega).$$

This formula applies the Gram-Schmidt orthogonalization to eliminate components of v in the directions of active constraint gradients, ensuring constraint adherence under the corrected dynamics. Under LICQ, this projection is well-defined and continuous in ω .

8.3 Equilibria and Fixed Points

Definition 8.2 (Equilibrium)

A point $\omega^* \in \mathcal{M}$ is an *equilibrium* if there exists $u^* \in U$ such that $f(\omega^*, u^*) = 0$. The governed manifold is required to contain at least one stable equilibrium $\omega^* \in \Sigma$, called the *principal alignment point*.

Lyapunov Stability. An equilibrium $\omega^* \in \mathcal{M}$ is *Lyapunov stable* if for every $\varepsilon > 0$ there exists $\delta > 0$ such that $\|\omega(0) - \omega^*\|_\Omega < \delta$ implies $\|\omega(t) - \omega^*\|_\Omega < \varepsilon$ for all $t \geq 0$.

Asymptotic Stability. An equilibrium $\omega^* \in \mathcal{M}$ is *asymptotically stable* if it is Lyapunov stable and additionally $\|\omega(t) - \omega^*\|_\Omega \rightarrow 0$ as $t \rightarrow \infty$ for all trajectories with initial condition sufficiently near ω^* .

The principal requires the principal alignment point $\omega^* \in \Sigma$ to be asymptotically stable within $\mathcal{M}$.

8.4 Flow Map

Definition 8.3 (Flow Map)

The *flow map* is $\Phi^t: \mathcal{M} \rightarrow \mathcal{M}$, defined by $\Phi^t(\omega_0) = \omega(t)$ where ω is the unique solution to the governed ODE with initial condition $\omega(0) = \omega_0$. The family $\{\Phi^t\}_{t \geq 0}$ is a one-parameter semigroup: $\Phi^0 = id$ and $\Phi^{s+t} = \Phi^s \circ \Phi^t$.

The central safety requirement is the *positive invariance* of $\mathcal{M}$ under the flow:

$$\Phi^t(\mathcal{M}) \subseteq \mathcal{M} \quad \forall t \geq 0.$$

This is guaranteed by the Nagumo condition (Definition 4.5) and the existence of a barrier certificate (Definition 4.3). Under Lipschitz regularity of f , the flow map Φ^t is Lipschitz in ω_0 with constant e^{Lt} (where L is the Lipschitz constant of f), by Gronwall's inequality.

9 Formal Governance Conditions

9.1 Completeness

Definition 9.1 (Completeness)

The invariant family $\{I_k\}$ is *complete* if for any inadmissible behavior $\omega \notin \mathcal{M}$, there exists at least one $k \in \{1, \dots, K\}$ such that $I_k(\omega) = 0$. Equivalently, the invariant family covers all unsafe behaviors: $\Omega \setminus \mathcal{M} \subseteq \cup_k \{I_k = 0\}$.

9.2 Consistency

Definition 9.2 (Consistency)

The invariant family is *consistent* if $\mathcal{M} \neq \emptyset$: there exists at least one state-action pair satisfying all governance constraints simultaneously. Inconsistency is a fatal specification error.

9.3 Minimal Sufficient Invariant Set

Definition 9.3 (Minimal Sufficient Invariant Set)

A subset $\{I_{kj}\}$ is a *minimal sufficient invariant set* if it is (i) consistent, (ii) complete, and (iii) no proper subset is both consistent and complete. The minimal sufficient invariant set is the irredundant basis C^* of Definition 7.3.

9.4 Principal Monotonicity

Definition 9.4 (Principal Monotonicity)

The invariant family is *principal-monotone* if tightening any constraint (i.e., replacing φ_k by $\varphi_k - \varepsilon$ for any $\varepsilon > 0$) strictly shrinks the governed manifold and does not create new feasible regions. Formally, if $\mathcal{M}_\varepsilon = \mathcal{M} \cap \{\varphi_k \leq -\varepsilon\}$, then $\mathcal{M}_\varepsilon \subsetneq \mathcal{M}$ for all $\varepsilon > 0$. This property ensures that the principal retains monotone control over the feasible region.

9.5 Governance Theorem

Theorem 9.1 (Governed Manifold Invariance — Main Theorem)

Suppose the following hypotheses hold:

1. The invariant family $\{I_k\}$ is complete and consistent (Definitions 9.1 and 9.2).
2. The governed manifold $\mathcal{M}$ is positively invariant under the transition operator T .
3. The Nagumo condition holds on $\partial\mathcal{M}$ (Definition 4.5).
4. A safety Lyapunov function V exists for the safety envelope Σ (Definition 6.2).

Conclusion: All admissible trajectories originating in Σ remain in Σ for all $t \geq 0$. All admissible trajectories originating in $\mathcal{M}$ remain in $\mathcal{M}$ for all $t \geq 0$. No admissible trajectory can enter the forbidden zone $F = \Omega \setminus \mathcal{M}$.

Proof. Invariance of $\mathcal{M}$: By hypotheses (2) and (3), together with the barrier certificate of Definition 4.3, no trajectory initiated in $\mathcal{M}$ can exit $\mathcal{M}$; hence F is unreachable. Invariance of Σ : By Theorem 6.1 (Lyapunov Safety) and hypothesis (4), all trajectories in Σ remain in Σ . Completeness (hypothesis 1) ensures that $\mathcal{M}$ contains no inadmissible behavior, so governance is total. $\square$

10 Notation and Symbol Glossary

Symbol	Meaning	Type	Domain / Range
$\mathcal{S}$	State space	Banach space	$\mathbb{R}^n$
$\mathcal{A}$	Action space	Compact metric space	—
Ω	Product space $\mathcal{S} \times \mathcal{A}$	Topological space	$\mathcal{S} \times \mathcal{A}$
s	State vector	Vector	$\mathcal{S}$
a	Action	Point	$\mathcal{A}$
ω	State-action pair (s, a)	Point	Ω
I_k	k-th invariant predicate	Boolean function	$\Omega \rightarrow \{0, 1\}$
φ_k	k-th constraint function	Real function (Lipschitz)	$\Omega \rightarrow \mathbb{R}$
K	Number of invariants	Integer	$\mathbb{N}$
I_{total}	Total invariant predicate	Boolean function	$\Omega \rightarrow \{0, 1\}$
Φ	Aggregate constraint function	Real function (Lipschitz)	$\Omega \rightarrow \mathbb{R}$
$\mathcal{M}$	Governed manifold	Closed subset	Ω
$\mathcal{M}^\circ$	Interior of $\mathcal{M}$	Open subset	Ω
$\partial\mathcal{M}$	Boundary of $\mathcal{M}$	Closed subset	Ω
Σ	Safety envelope	Subset	$\mathcal{M}$
δ_{safety}	Safety margin	Scalar	$\mathbb{R}_{>0}$
$\mathcal{T}$	Transition relation	Relation	$\mathcal{M} \times \mathcal{M}$
B	Barrier / safety potential function	Real function	$\mathcal{M}^\circ \rightarrow \mathbb{R}$

Symbol	Meaning	Type	Domain / Range
V	Safety Lyapunov function	Real function	$\Omega \rightarrow \mathbb{R}_{\geq 0}$
f	Governing vector field	Vector field	$\mathcal{M} \times U \rightarrow T\Omega$
Φ^t	Flow map at time t	Map	$\mathcal{M} \rightarrow \mathcal{M}$
∇	Gradient operator	Operator	—
$T_\omega \mathcal{M}$	Tangent space / cone at ω	Linear space / convex cone	—
$N_\omega \mathcal{M}$	Normal cone at ω	Convex cone	—
$d_{\mathcal{M}}$	Geodesic distance on $\mathcal{M}$	Metric	$\mathcal{M} \times \mathcal{M} \rightarrow \mathbb{R}_{\geq 0}$
r_{max}	Maximum state norm	Scalar	$\mathbb{R}_{>0}$
κ	KL-divergence threshold	Scalar	$\mathbb{R}_{>0}$
ρ_{max}	Irreversibility threshold	Scalar	$(0,1)$
R_{max}	Resource budget	Scalar	$\mathbb{R}_{>0}$
τ_{max}	Maximum step size	Scalar	$\mathbb{R}_{>0}$
η	Disclosure tolerance	Scalar	$\mathbb{R}_{\geq 0}$
G	Constraint dependency graph	Undirected graph	$V \times E$
Ξ	Constraint sensitivity matrix	Matrix	$\mathbb{R}^{K \times n}$
Irr	Irreversibility measure	Real function	$\Omega \rightarrow [0,1]$
D_{KL}	Kullback-Leibler divergence	Functional	$\mathbb{R}_{\geq 0}$
$A_{forbidden}$	Set of prohibited actions	Closed subset	A
U	Control input space	Compact metric space	—

Symbol	Meaning	Type	Domain / Range
C^*	Irredundant constraint basis	Index set	$\{1, \dots, K\}$

Appendix A — Constraint Function Specifications

This appendix provides the standard algebraic form, parameterization, and tuning guidance for each of the ten canonical constraint functions.

A.1 Standard Form Table

ID	Standard Form $\varphi_k(s, a)$	Parameters	Tuning Guidance
I_1	$\ s\ _S - r_{max}$	$r_{max} > 0$	Set to the operational state bound. Larger r_{max} = more permissive.
I_2	$d_A(a, A_{forbidden}) - \varepsilon$	$A_{forbidden}, \varepsilon \geq 0$	$\varepsilon > 0$ provides a safety buffer around forbidden actions. $A_{forbidden}$ must be specified by the principal.
I_3	$\ \nabla_s V(s)\ - g_{max}$	$V: S \rightarrow \mathbb{R}, g_{max} > 0$	Choose V as the principal's value function. g_{max} bounds gradient magnitude; set conservatively.
I_4	$\max_{t' < t} [\ell(s_t, s_{t'}, a_{t'}) - \delta]$	$\ell, \delta > 0$	ℓ encodes the causal model; δ is the causal tolerance. Application-specific design required.
I_5	$D_{KL}(\pi_{agent} \parallel \pi_{principal}) - \kappa$	$\pi_{principal}, \kappa > 0$	Small κ enforces tight alignment. $\pi_{principal}$ must be

ID	Standard Form $\varphi_k(s, a)$	Parameters	Tuning Guidance
			updated as principal intent evolves.
I_6	$R(s, a) - R_{max}$	$R: \Omega \rightarrow \mathbb{R}_{\geq 0}, R_{max} > 0$	Define R to include compute, energy, monetary cost, or any metered resource. R_{max} is the operational budget.
I_7	$Irr(s, a) - \rho_{max}$	$Irr: \Omega \rightarrow [0, 1], \rho_{max} \in (0, 1)$	Set ρ_{max} near 0 for maximum caution. Irr must be estimated from transition dynamics.
I_8	$\ H_{agent}(s) - H_{disclosed}(s)\ - \eta$	$\ \cdot\ , \eta \geq 0$	Choose $\ \cdot\ $ as total variation or KL divergence. $\eta = 0$ enforces full transparency.
I_9	$\ \omega_t - \omega_{t-1}\ _{\Omega} - \tau_{max}$	$\tau_{max} > 0$	Set τ_{max} to bound the maximum velocity of the trajectory. Smaller values enforce smoother dynamics.
I_{10}	$[scope(a) \notin S_{permitted}]$ as $\{-1, +1\}$ -valued	$scope: A \rightarrow \mathcal{P}(\text{Domain}), S_{permitted}$	Maintain a lattice of permitted scopes. Update $S_{permitted}$ upon explicit principal authorization only.

A.2 Parameter Sensitivity

Let $\theta = (r_{max}, \varepsilon, g_{max}, \delta, \kappa, R_{max}, \rho_{max}, \eta, \tau_{max})$ denote the vector of tunable scalar parameters. The governed manifold is a monotone decreasing function of θ in the sense that decreasing any component of θ weakly shrinks $\mathcal{M}$ (principal

monotonicity, Definition 9.4). The sensitivity of $\text{vol}(\mathcal{M})$ to θ_i can be estimated numerically via Monte Carlo volume estimation or analytically via the coarea formula where φ_k is smooth.

Appendix B — Existence and Uniqueness Conditions

This appendix states conditions under which the Cauchy problem for the governed dynamics admits a unique solution remaining in $\mathcal{M}$, extending the classical Picard-Lindelöf theorem to the constrained setting.

B.1 Setup

Consider the initial value problem:

$$d\omega/dt = f(\omega(t), u(t)), \quad \omega(0) = \omega_0 \in \mathcal{M},$$

where $u: \mathbb{R}_{\geq 0} \rightarrow U$ is a measurable control input and $f: \mathcal{M} \times U \rightarrow T\Omega$ is the governing vector field.

Theorem B.1 (Constrained Picard-Lindelöf)

Suppose:

1. $f(\cdot, u)$ is Lipschitz on $\mathcal{M}$ uniformly in $u \in U$ with constant $L > 0$.
2. $f(\omega, \cdot)$ is continuous on U for each $\omega \in \mathcal{M}$.
3. $u(\cdot)$ is measurable and essentially bounded.
4. The Nagumo condition holds: $f(\omega, u) \in T_\omega \mathcal{M}$ for all $\omega \in \partial \mathcal{M}$, all $u \in U$.

Conclusion: For every $\omega_0 \in \mathcal{M}$, the Cauchy problem has a unique absolutely continuous solution $\omega: [0, \infty) \rightarrow \mathcal{M}$. The solution map $\omega_0 \mapsto \omega(t)$ is Lipschitz in ω_0 with constant e^{Lt} (Gronwall estimate).

Proof sketch. Existence and uniqueness on $[0, T_{\max})$ follow from the standard Carathéodory extension of Picard-Lindelöf under Lipschitz regularity. Global

existence ($T_{\max} = \infty$) follows from the compactness of $\mathcal{M}$ (Proposition 3.4): the trajectory cannot blow up in finite time since $\mathcal{M}$ is bounded. Invariance of $\mathcal{M}$ follows from condition (4) via the Nagumo-Bony theorem. $\square$

B.2 Regularity of the Flow Map

Under the hypotheses of Theorem B.1, the flow map $\Phi^t: \mathcal{M} \rightarrow \mathcal{M}$ (Definition 8.3) is:

- Lipschitz continuous in (ω_0, t) .
- Jointly measurable in (ω_0, t, u) .
- Differentiable in ω_0 at interior points if f is C^1 , with the variational equation governing the Jacobian $\partial\Phi^t/\partial\omega_0$.

If f is C^k on $\mathcal{M}^\circ$, then Φ^t is C^k on the interior, by the classical smooth dependence theorem for ODEs.

Appendix C — Numerical Implementation Notes

C.1 Approximation of the Governed Manifold

In practice, $\mathcal{M}$ is approximated by evaluating Φ on a finite grid or via sampling. The following procedures are recommended:

- 12. Grid-Based Approximation.** Discretize S on a uniform grid of resolution $h > 0$. For each grid point s_i and action $a_j \in A$ (discretized), evaluate $\Phi(s_i, a_j)$ and retain points with $\Phi \leq 0$. The approximation error is $O(h \cdot L)$ in Hausdorff distance, where L is the Lipschitz constant of Φ .
- 13. Sample-Based Approximation.** Draw N samples uniformly from Ω and retain those satisfying $\Phi(\omega) \leq 0$. The empirical approximation converges in distribution to $\text{cal } M$ as $N \rightarrow \infty$ by the Glivenko-Cantelli theorem.
- 14. Level-Set Methods.** Represent $\text{cal } M$ implicitly as the zero sublevel set of Φ , evolving Φ numerically under the Hamilton-Jacobi equation for reachability analysis.

C.2 Evaluation of the Aggregate Constraint Function

Evaluating $\Phi(\omega) = \max_k \phi_k(\omega)$ requires evaluating all K constraint functions. For the canonical $K = 10$ system:

- L_1 : $O(n)$ via norm computation.
- L_2 : $O(|A_{\text{forbidden}}|)$ via nearest-neighbor search.
- L_3 : $O(n)$ via automatic differentiation of V .
- L_5 : $O(|A|)$ via numerical KL divergence integration.
- All others: $O(1)$ or $O(n)$ via direct computation.

Total evaluation cost is $O(K \cdot n + |A_{\text{forbidden}}|)$ per point.

C.3 Projection onto the Tangent Cone

To project a velocity vector $v \in T_\omega \Omega$ onto $T_\omega \mathcal{M}$ at a boundary point $\omega \in \partial \mathcal{M}$:

15. Identify the active constraint set $\mathcal{A} = \{k : \phi_k(\omega) = 0\}$.
16. Form the matrix $J \in \mathbb{R}^{|\mathcal{A}| \times (n + \dim A)}$ with rows $\nabla \phi_k(\omega)$ for $k \in \mathcal{A}$.
17. Solve the projection: $\text{Proj}_{T_\omega \mathcal{M}}(v) = v - J^{\top} (J J^{\top})^{-1} J v$ via QR decomposition of J .

Cost: $O(|\mathcal{A}|^2 \cdot (n + \dim A))$ per projection. For $|\mathcal{A}| \ll K$, this is efficient in practice.

C.4 Computation of the Safety Potential

The logarithmic barrier $B(\omega) = -\sum_k \log(-\phi_k(\omega))$ is evaluated in $O(K)$ operations per point. Its gradient

$$\nabla B(\omega) = \sum_{k=1}^K \nabla \phi_k(\omega) / (-\phi_k(\omega))$$

is a repulsive vector field pointing away from the boundary, suitable as a penalty term in trajectory optimization objectives. Numerical conditioning degrades as $\omega \rightarrow \partial \mathcal{M}$ (since $-\phi_k(\omega) \rightarrow 0$); a minimum slack threshold $\epsilon_{\min} > 0$ should be maintained in implementations to avoid division by

values near zero. This is precisely the role of δ_{safe} : confining computation to Σ ensures $\|\phi_k(\omega)\| \geq \delta_{\text{safe}} > 0$ uniformly.

Governed Manifold Specification — Version 1.0 — Leon (Principal) — October 2026

This document constitutes the canonical formal specification of the Autonomy Substrate. All governed systems are bound by the definitions, propositions, and theorems herein.